

Chapter 2 – Semiconductor Device Physics Fundamentals

- 2.1 Semiconductor Materials
- 2.2 Crystal Structure
- 2.3 Semiconductor Models
- 2.4 Semiconductor Doping
- 2.5 Carrier Statistic
- 2.6 Carrier Transport
- 2.7 Carrier Generation/Recombination

Literature:

Pierret, Chapter 1-3, page 1-132

Sze, Chapter 1-3, page 1-83

2.1 Semiconductor Materials

Material Classification by **Resistivity**:

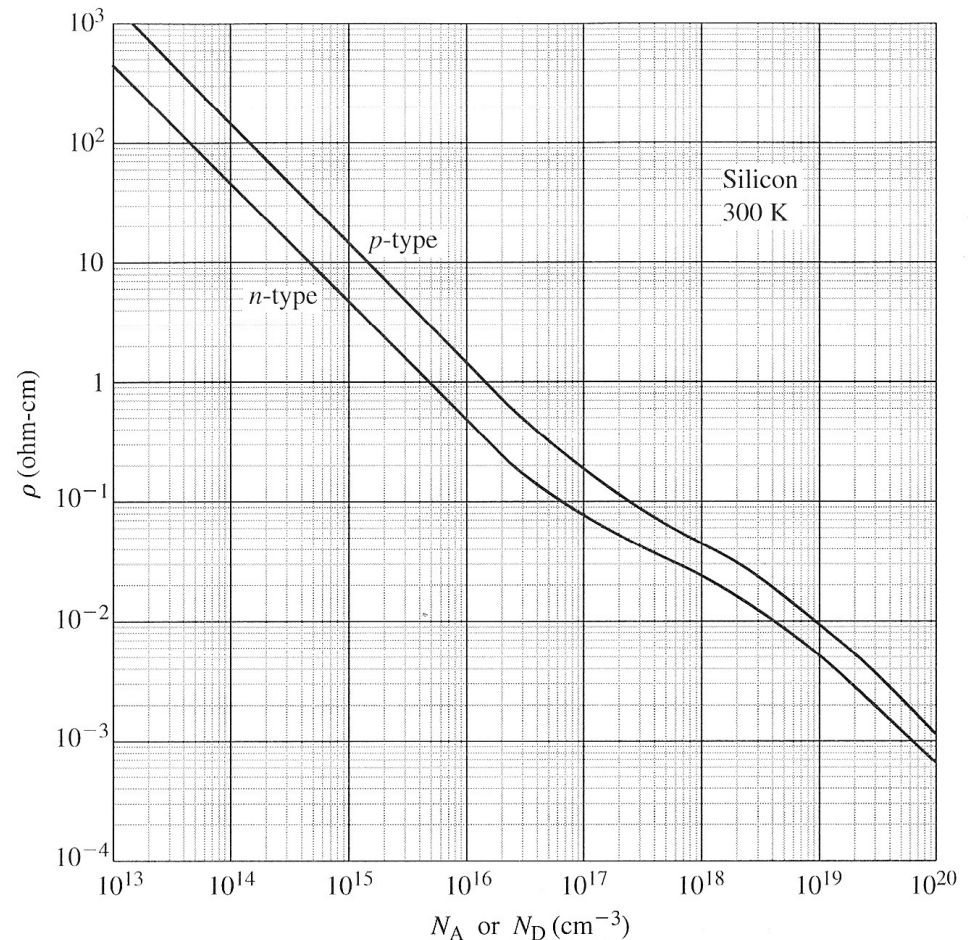
Material	Resistivity ρ [$\Omega \cdot \text{cm}$]
Insulators	$10^5 < \rho$
Semiconductors	$10^{-3} < \rho < 10^5 (-10^8)$
Conductors	$\rho < 10^{-3}$

Jaeger, Blalock, Table 2.1

Literature: Pierret, Chapter 1.1, page 3-6

Semiconductor Materials

- Semiconductors have conductivities between those of insulators and those of conductors
- Semiconductors are materials whose **electric properties** (e.g. conductivity & resistivity) can be **controlled** over a wide range **by doping**, i.e. addition of controlled amounts of specific impurity atoms
- Semiconductor materials are the base of all semiconductor devices, such as diodes and transistors (BJT and MOSFET)



Pierret, Fig. 3.8

Resistivity/Conductivity of Insulators, Semiconductor and Conductors

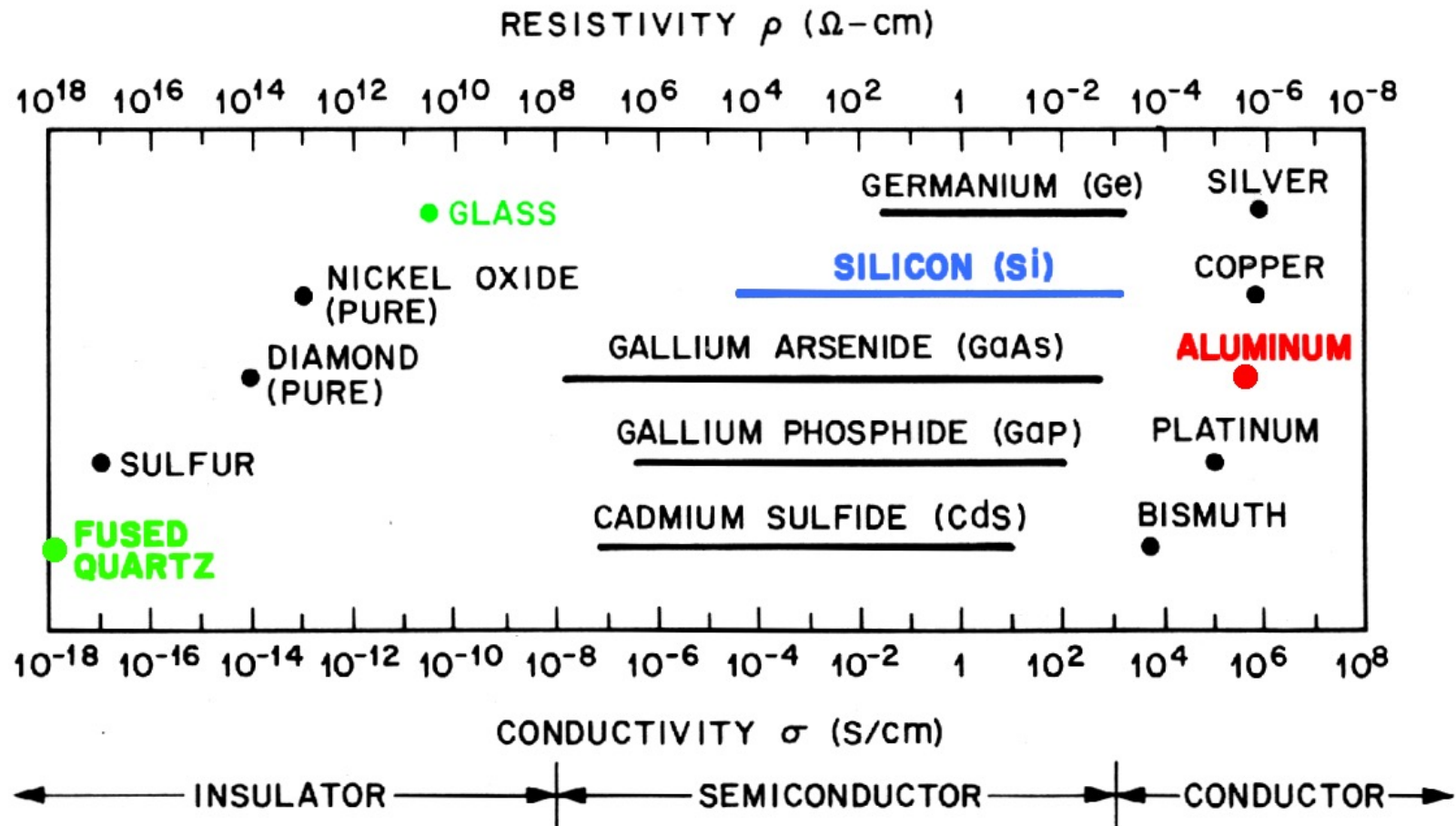


Fig. 1 Typical range of conductivities for insulators, semiconductors, and conductors.

Sze, Fig. 2.1

What Elements are Semiconductors based on?

- **Elemental Semiconductors:**
Silicon (Si), Germanium (Ge)
- **Compound Semiconductors:**
III-V: e.g. Gallium Arsenide (GaAs)
II-VI: e.g. Zinc Selenide (ZnSe)
IV-IV: Silicon Carbide (SiC)
- **Alloy Semiconductors:**
Binary: $\text{Si}_{1-x}\text{Ge}_x$
Ternary: $\text{Al}_x\text{Ga}_{1-x}\text{As}$
 $\text{Al}_x\text{In}_{1-x}\text{As}$
 $\text{GaAs}_{1-x}\text{P}_x$
Quaternary: $\text{Ga}_x\text{In}_{1-x}\text{As}_{1-y}\text{P}_y$
- Because of its well-developed fabrication technology, **SILICON** is by far the most important semiconductor material

	IIIA	IVA	VA	VIA
	5 10.811 B Boron	6 12.01115 C Carbon	7 14.0067 N Nitrogen	8 15.9994 O Oxygen
	13 26.9815 Al Aluminum	14 28.086 Si Silicon	15 30.9738 P Phosphorus	16 32.064 S Sulfur
IIIB	30 65.37 Zn Zinc	31 69.72 Ga Gallium	32 72.59 Ge Germanium	33 74.922 As Arsenic
	48 112.40 Cd Cadmium	49 114.82 In Indium	50 118.69 Sn Tin	51 121.75 Sb Antimony
	80 200.59 Hg Mercury	81 204.37 Tl Thallium	82 207.19 Pb Lead	83 208.980 Bi Bismuth
				84 (210) Po Polonium

Shaded = most important semiconductor elements

Jaeger, Blalock, Table 2.2

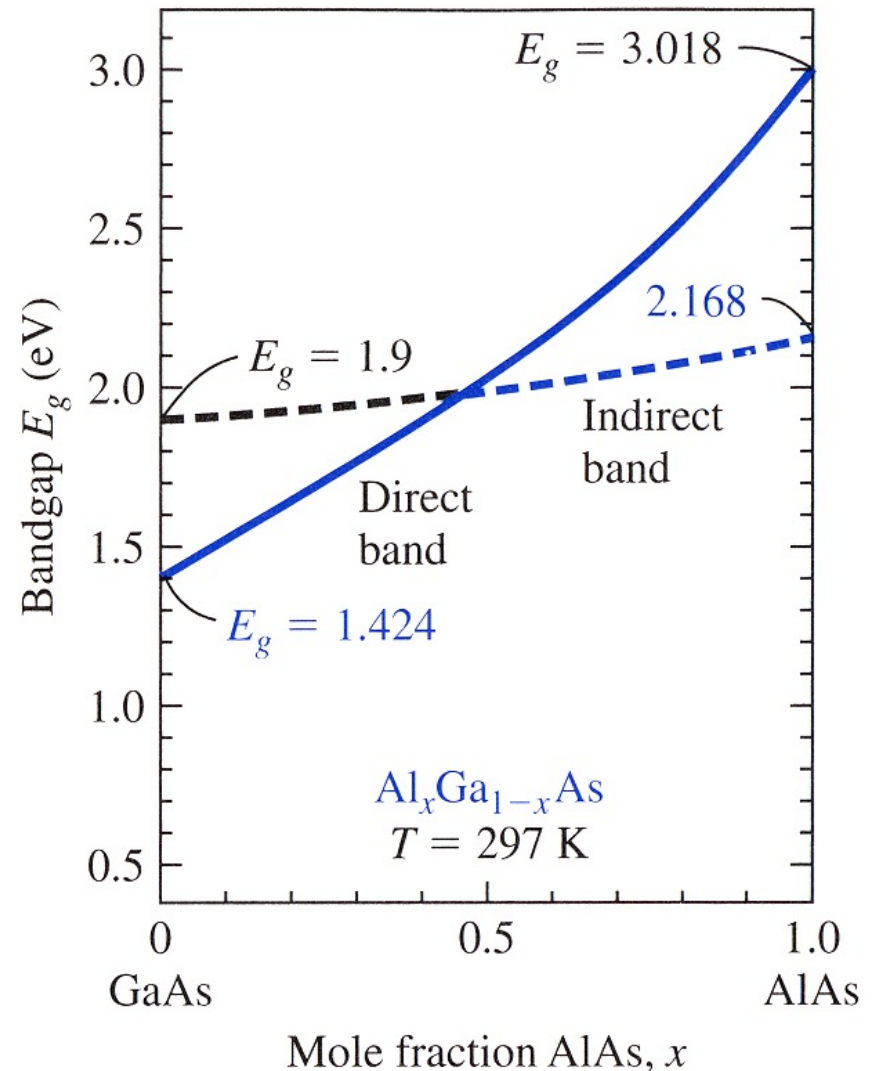
Element and Compound Semiconductors

Table 2 Element and Compound Semiconductors

Element	IV–IV Compounds	III–V Compounds	II–VI Compounds	IV–VI Compounds
Si	SiC	AlAs	CdS	PbS
Ge		AlSb	CdSe	PbTe
		BN	CdTe	
		GaAs	ZnS	
		GaP	ZnSe	
		GaSb	ZnTe	
		InAs		
		InP		
		InSb		

The “Other” Semiconductor Materials

- Silicon is the dominating semiconductor material
- The “other” semiconductor materials are utilized in applications requiring e.g.
 - High speed
 - Optoelectronic properties
 - High temperature operation
- Example: Ternary alloys, such as $\text{Al}_x\text{Ga}_{1-x}\text{As}$, have a “tunable” bandgap, enabling LEDs and laser diodes with “engineered” output spectrum
- Substantial research is performed (also at GaTech!) on both Si and non-Si semiconductor devices



2.2 Crystal Structure

2.2.1 Unit Cell

- Unit Cell & Primitive Unit Cell
- Cubic Crystal Structure

2.2.2 Semiconductor Lattice

- Diamond lattice
- Zincblende lattice

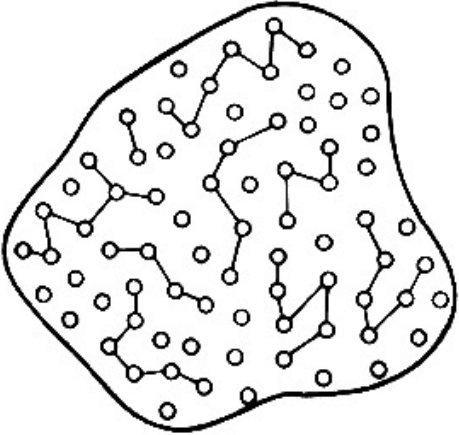
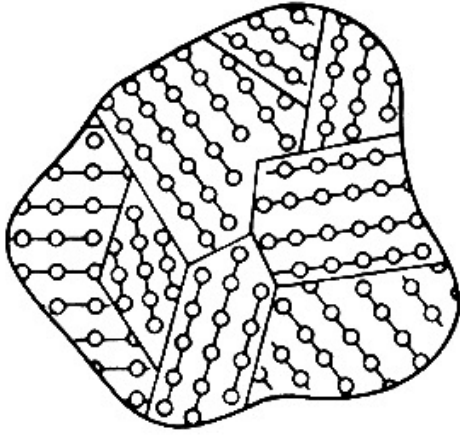
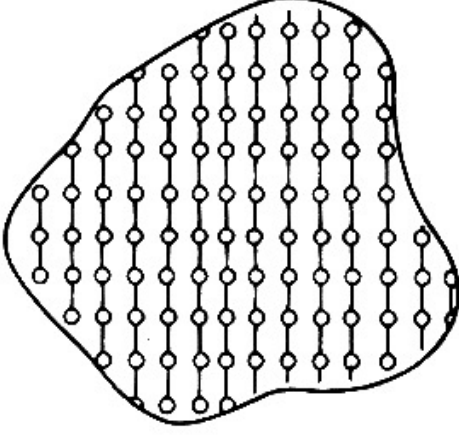
2.2.3 Miller Indices

- Crystal Directions
- Crystal Planes

Literature: Pierret, Chapter 1.2, page 6-16

Classification of Solids

Solids can be classified based on their degree of atomic order

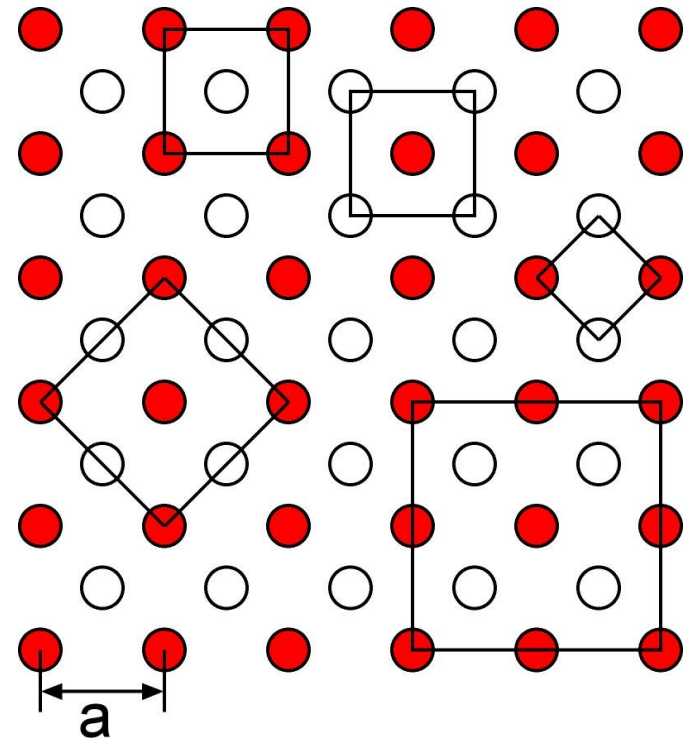
		
Amorphous No recognizable long-range order	Polycrystalline Solid is made up of crystallites, i.e. segments which are completely ordered	Crystalline Entire solid is made up of atoms in an orderly array

Pierret, Fig. 1.1

Most semicon. devices are based on **crystalline semiconductors!**

2.2.1 Crystal Unit Cell

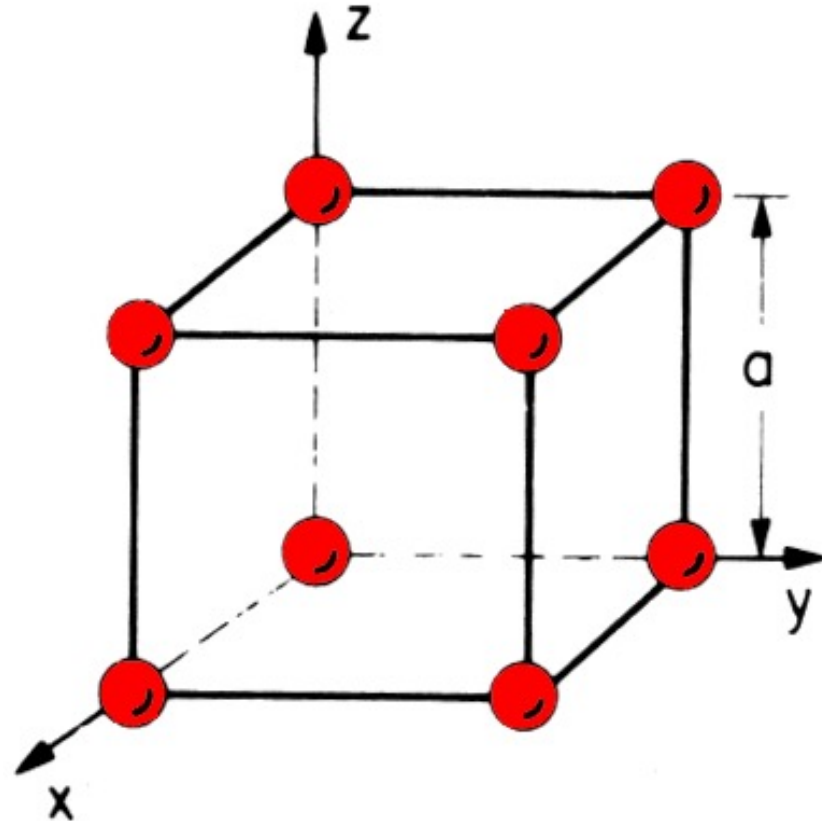
- **Crystal** = Periodic 3-D arrangement of atoms
- **Unit cell** = small portion of the crystal that can be used to reproduce the crystal
- **Primitive (unit) cell** = smallest unit cell possible
- Unit cells (including primitive unit cells) are not unique
- A unit cell does not need to be primitive; often it is more convenient to have a slightly larger unit cell with orthogonal sides (instead of primitive cell with non-orthogonal sides)
- WWW-page on crystal structures:
http://departments.kings.edu/chemlab/chemlab_v2/



*Which cells are unit cells?
Which cells are primitive cells?
Which cell is not a unit cell?*

Cubic Crystal Structure

- Simple cubic (sc) is the simplest of all 3-D unit cells:
 - Cube with an atom positioned at each corner
 - Only 1/8 of each corner atom is inside the unit cell: 1 atom per unit cell; each atom has 6 nearest neighbors
 - The side length is called the lattice constant a
- Very few materials crystallize in a simple-cubic lattice
- Example: α -polonium
- How can we “see” the crystal structure of a crystal?



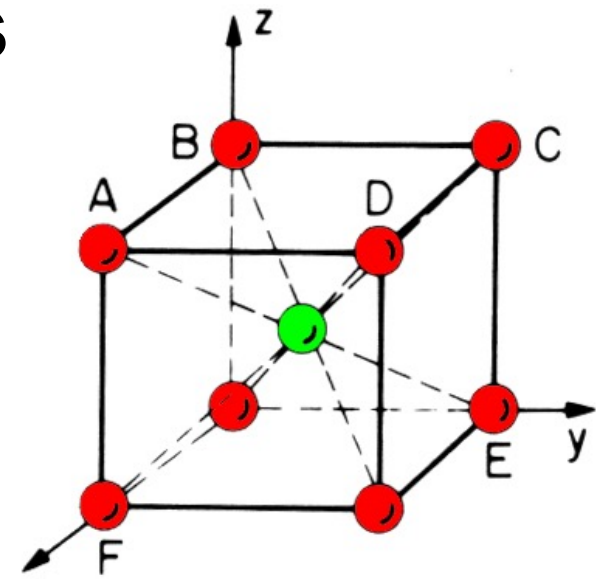
Sze, Fig. 2.3

Cubic Crystal Structures

– *fcc* and *bcc* Lattice –

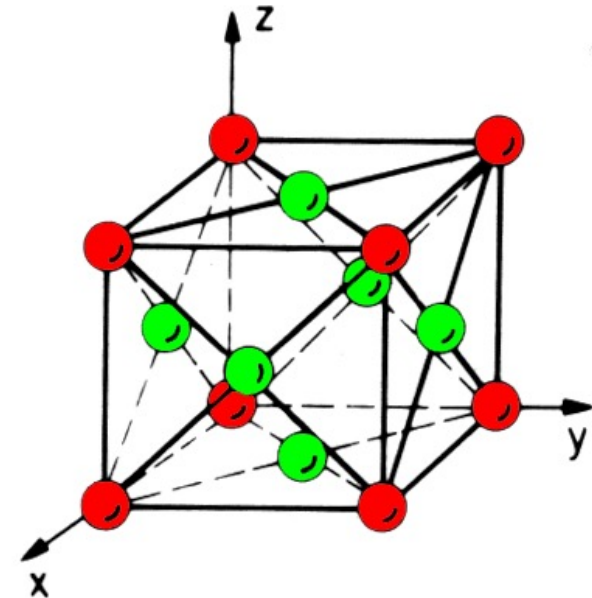
Body-Centered-Cubic (bcc) unit cell

- Additional atom in the center of the cube
- 2 atoms per unit cell
8 nearest neighbors for each atom
- Examples: **Cr, W, Na, Fe**



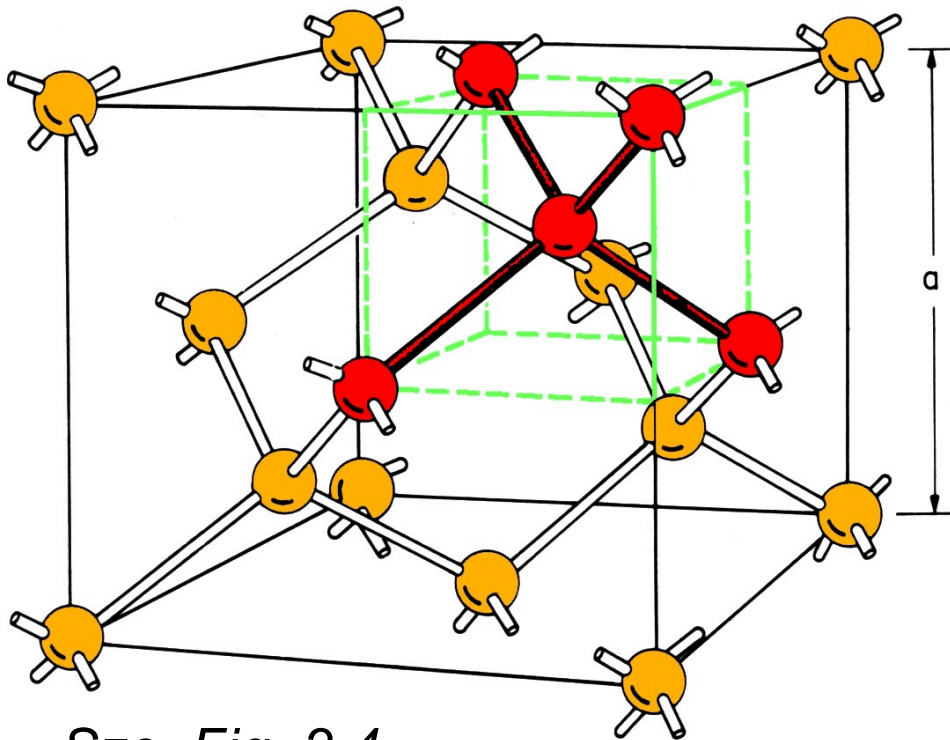
Face-Centered-Cubic (fcc) unit cell

- Additional atoms in the center of each face of the cube
- Also known as cubic-close-packed (ccp)
- 4 atoms per unit cell
12 nearest neighbors for each atom
- Examples: **Ni, Ag, Au, Cu, Al**



Sze, Fig. 2.3

2.2.2 Diamond Lattice Silicon Crystal Structure

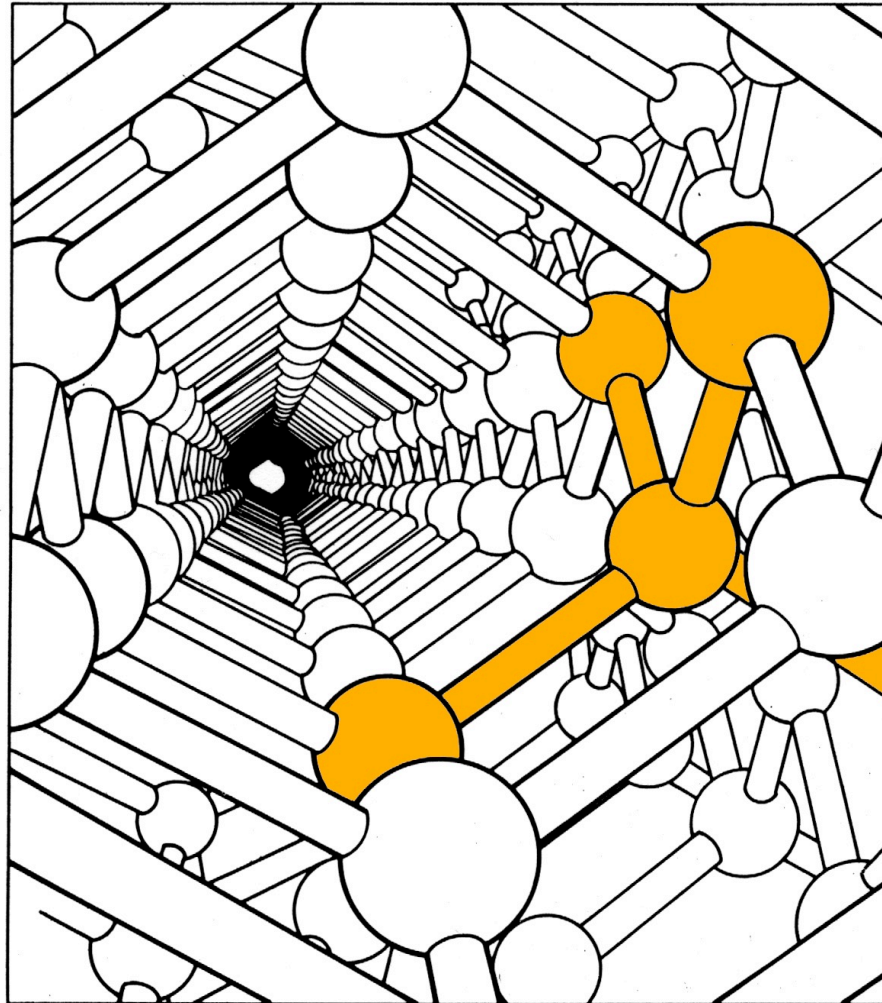


Sze, Fig. 2.4

Calculate: How many Si atoms do we have per cm^{-3} ?

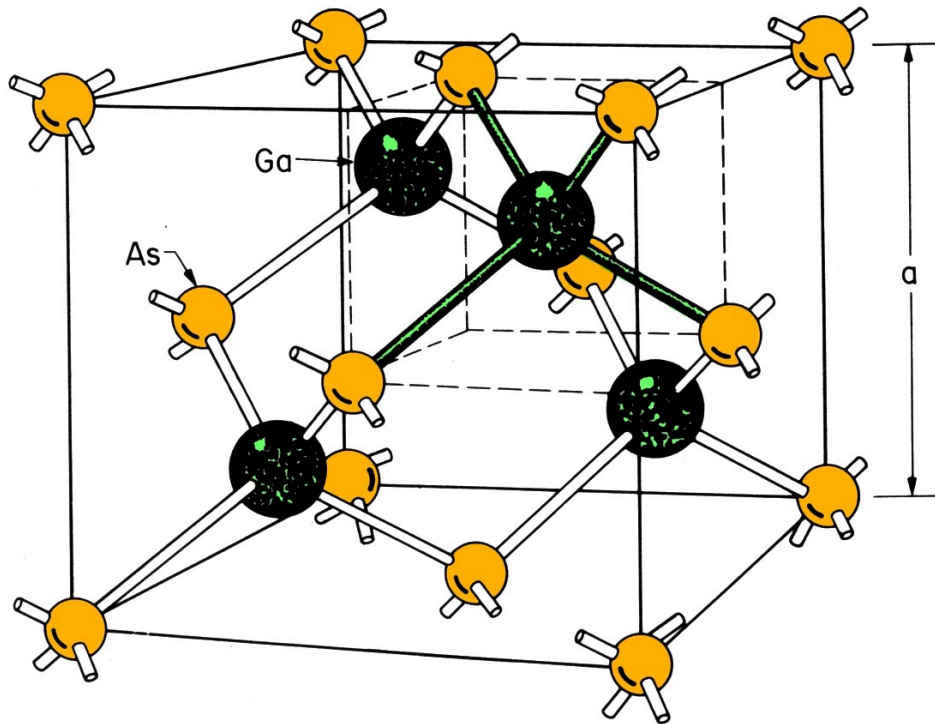
- Silicon and germanium have a diamond lattice, a special cubic crystal structure
- The diamond lattice consists of two interpenetrating face-centered-cubic (fcc) lattices shifted by one-quarter of the cube body diagonal
- Each silicon atom has 4 nearest neighbors forming a tetrahedral structure
- The silicon unit cell has a lattice constant $a = 5.43 \text{ \AA}$ ($T = 300 \text{ K}$) ($1 \text{ \AA} = 10^{-10} \text{ m} = 10^{-8} \text{ cm} = 0.1 \text{ nm}$)
- Each silicon atom forms four covalent bonds with its nearest neighbors (see Chapter 2.3)

Diamond Lattice along $\langle 110 \rangle$ Axis



Zincblende Lattice

GaAs Crystal Structure

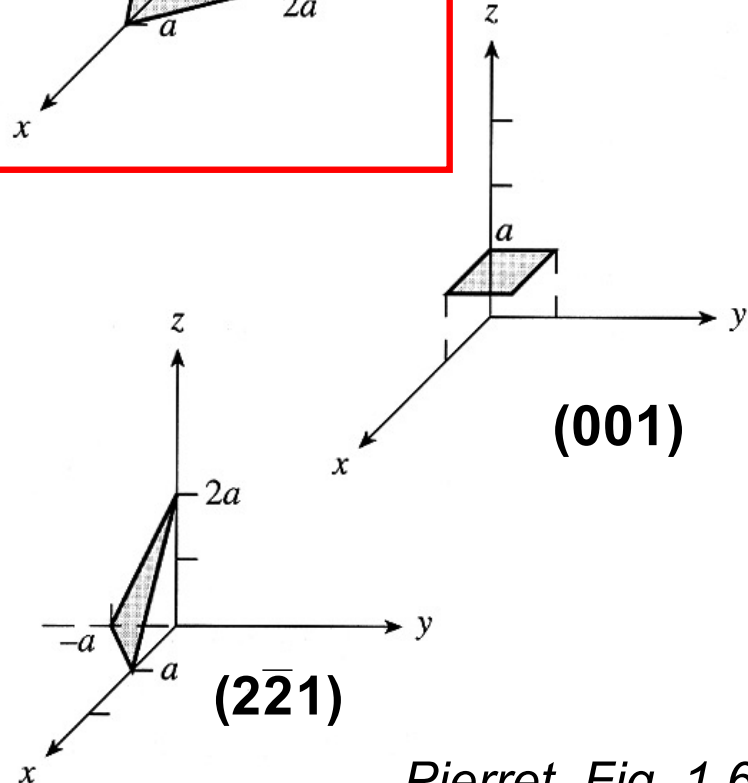
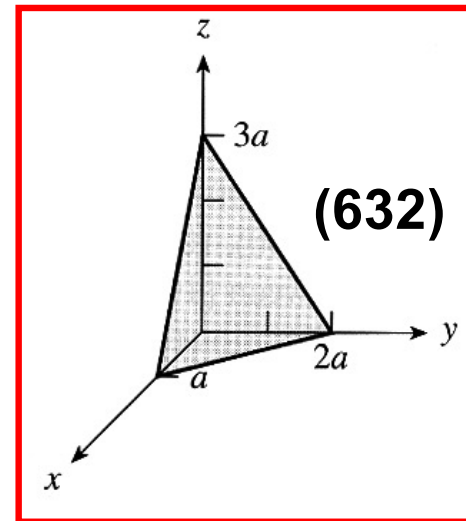


- Most III-V semiconductors (e.g. GaAs) have a zincblende lattice, again a cubic crystal structure
- Similar to the diamond lattice, the zincblende lattice consists of two interpenetrating face-centered-cubic (fcc) lattices shifted by one-quarter of the cube body diagonal, but with different atoms occupying each fcc sublattice
- The GaAs unit cell has a lattice constant $a = 5.65 \text{ \AA}$ ($T = 300 \text{ K}$)
($1 \text{ \AA} = 10^{-10} \text{ m} = 10^{-8} \text{ cm} = 0.1 \text{ nm}$)

Sze, Fig. 2.4

2.2.3 Miller Indices

- Miller indices are used to **specify crystallographic planes and directions**
- How do I find the Miller index for a plane?
 - Record where the plane intersects coordinate axes in units of a : e.g. 1,2,3
 - Invert intercept values: e.g. 1, $1/2$, $1/3$
 - Multiply with constant to get smallest possible set of whole numbers: e.g. 6,3,2
 - Enclose set with curvilinear bracket: e.g. (632)
 - Note: $(6 - 32) \rightarrow (6\bar{3}2)$
- *What are the other plane's indices?*



Pierret, Fig. 1.6

More on Miller Indices

- Miller indices for directions are found analogous to the components of a vector
- In a cubic crystal, the planes

$$(100), (010), (001), (\bar{1}00), (0\bar{1}0), (00\bar{1}) \Rightarrow \{100\}$$

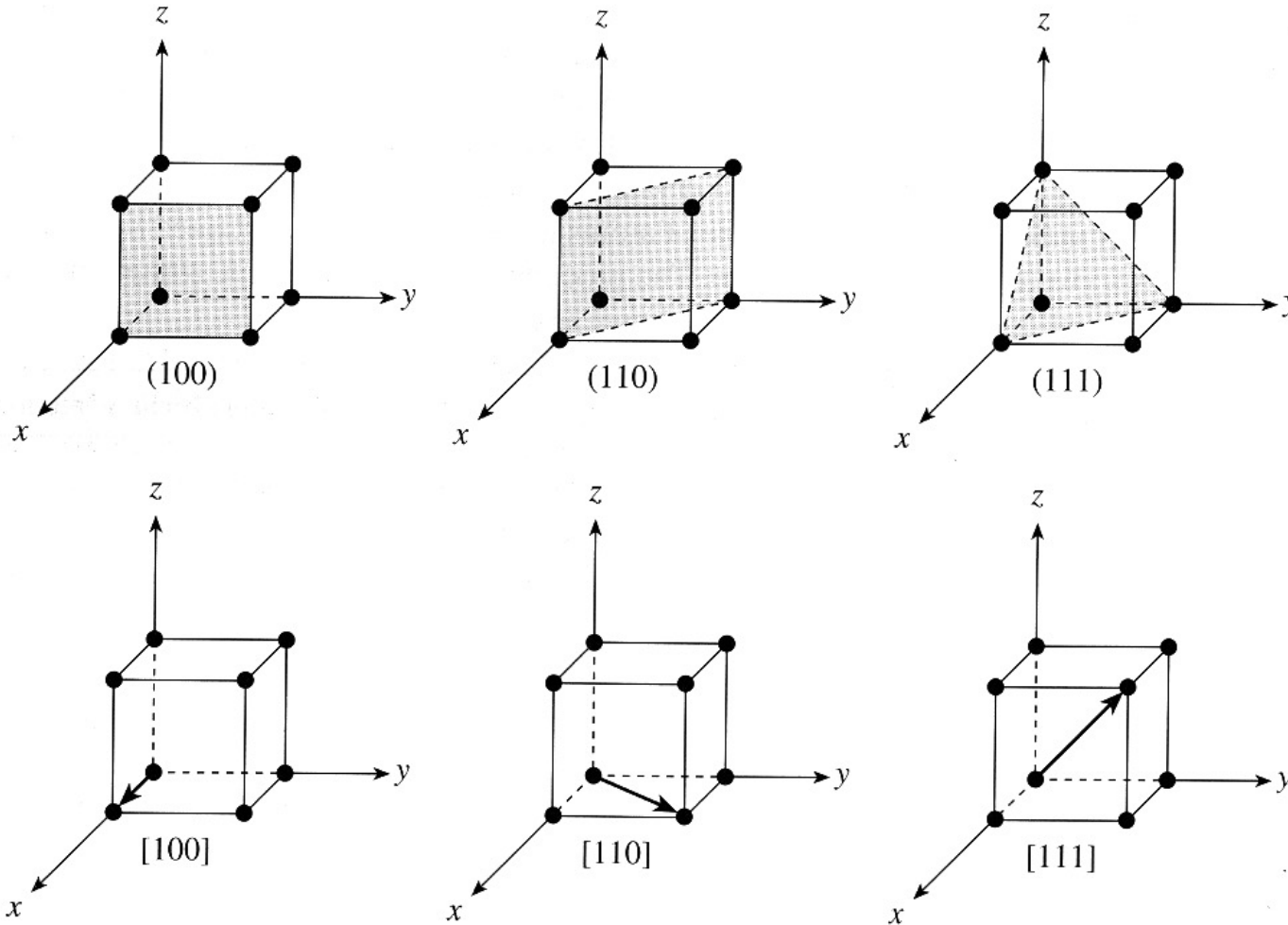
are **equivalent** because of symmetry and summarized as $\{100\}$

(hkl)	Particular crystal plane
$\{hkl\}$	Equivalent planes
$[hkl]$	Particular crystal direction
$\langle hkl \rangle$	Equivalent directions

- For cubic crystals, the direction $[hkl]$ is normal to the plane (hkl)

Crystal Planes and Directions

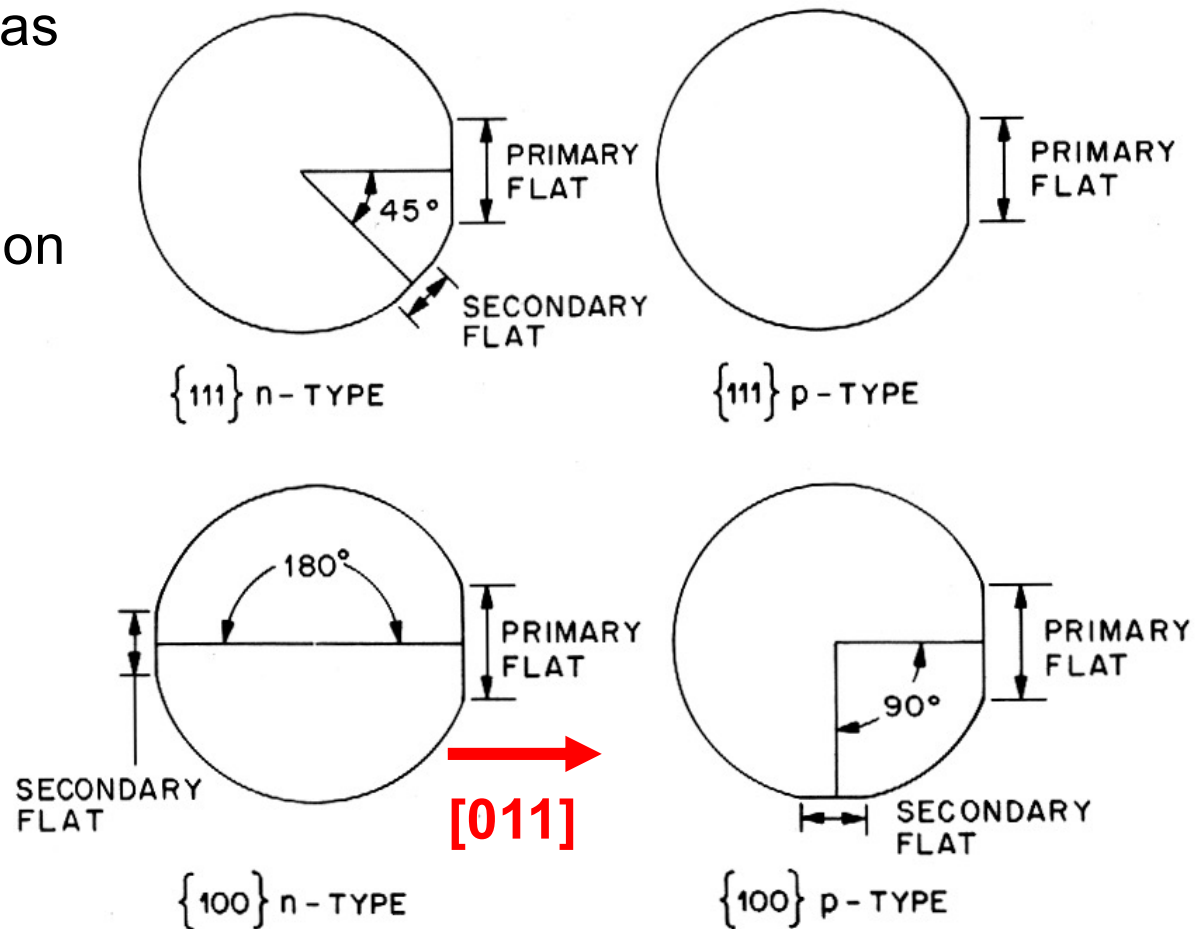
– Cubic Crystal Structure –



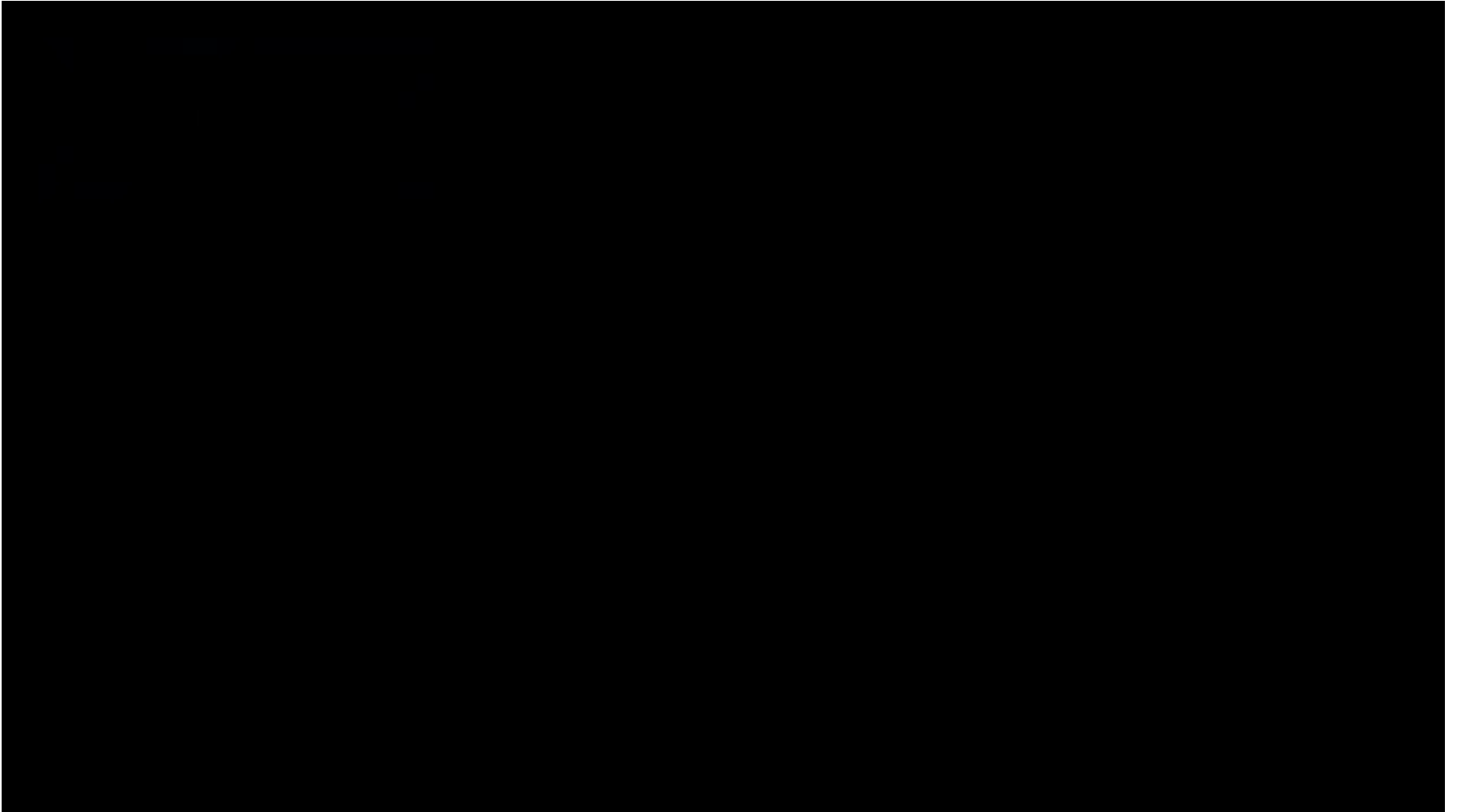
Pierret, Fig. 1.7

Wafer Flats

- Silicon wafers used for device processing have flats (or a notch) indicating crystal direction and doping type
- A “{100} wafer” has {100} surface
- For {100} wafer, the surface direction perpendicular to primary flat is [011]
- *How is a wafer fabricated?*



Silicon Wafer Production



<https://www.youtube.com/watch?v=AMgQ1-HdEIM>